# PEDAGOGY OF COMPUTER SCIENCE - I

**University Syllabus 2021-2022 onwards**

## UNIT - I : Aims and Objectives of Teaching Computer Science

Meaning, Nature, Scope, Need and Significance, Values,Aims and Objectives: Instructional objectives and Behavioural Objectives – Need and Importance of Instructional Objectives. Bloom's Taxonomy of Instructional Objectives: Cognitive, Affective and Psychomotor Domains, Revised Bloom's Taxonomy 2001 (Anderson & Krathwohl) Interrelation among the domains – Correlation between subjects.

## UNIT -II : Teaching Skills

Micro-Teaching : Concept, Definition, Steps, Cycle , Micro-teaching Vs Macro-Teaching - Skill of Set Induction - Skill of Explaining , Skill of Questioning , Probing skills, Skill of Stimulus Variation, Skill of Reinforcement, Skill of non-verbal clues, Skill of Closure - Link lesson – Model episode

## UNIT – III: Approaches of Teaching

Approaches of Lesson Planning - Steps - Organizing Teaching: Memory Level (Herbartian Model), Understanding Level (Morrison teaching Model), Reflective Level (Bigge and Hunt Teaching Model)– Unit Plan – Lesson Plan Writing.

## UNIT - IV : Methods Of Teaching

Teacher Centered Instruction: Lecture method, Demonstration and Team Teaching – Learner Centered Instruction: Self-Learning – Forms of Self-Learning: Programmed Instruction, Computer Assisted Instruction , Keller Plan, Project Method, Activity Based Learning (ABL), Active Learning Method (ALM)-Mind Map, Advanced Active Learning Method (AALM).

## Unit - V : Instructional Media

Classification of Instructional Media – Use of Mass media in classroom Instruction. New Emerging Media: Tele-Conferencing, Communication Satellites, Computer Networking, Word Processors, Blended Learning, Flipped Classroom, Artificial Intelligence, Augmented Reality.

# PEDAGOGY OF COMPUTER SCIENCE - I

## CONTENT

# UNIT - 1

# AIMS AND OBJECTIVES OF TEACHING COMPUTER SCIENCE

## 1.1 Computer Science : Meaning

Computer science is the scientific and practical approach to computation and its applications. It is the systematic study of the feasibility, structure, expression, and mechanization of the methodical procedures (or algorithms) that underlie the acquisition, representation, processing, storage, communication of, and access to information.

Computer science field can be divided into a variety of theoretical and practical disciplines. The computational complexity theory explores the fundamental properties of computational and intractable problems and they are highly abstract. The computer graphics emphasize real-world visual applications.

Computer science is studying the properties of computation. Software engineering is the design of specific computations to achieve practical goals.

Computer science is considered to have some closer relationship with mathematics than many scientific disciplines.

## 1.2 Nature of Computer Science

1. Access to expert and respected peers.
2. One to One and much communication.
3. Active learner participations.

4. Linking of new learning to concrete on the job problems.

5. Follow up, feedback and implementation support from pears or experts.

6. Self-direction control over stop or start, time, pace and place of learning or communication activity.

   - Computer Science Graduates Earn Higher Salaries
   - Computer Science Majors Have More Job Prospects
   - Computer Science Graduates Join a Fast Growing Industry
   - The software publishing industry is poised for growth (software publishing is a main focus of computer science degrees)
   - Computer Science Degree Holders Work in a Wide Range of Industries
   - Computer Science Graduates can Help Make the World a Better Place
   - Computer Science Graduates are on the Cutting-Edge of IT

## 1.3 Scope in computer science

1. **Software Developers**: Software developers are professionals who are concerned with facets of the software development process.

2. **Hardware Engineers**: These professionals do research, design, develop, test, and oversee the installation of computer hardware.

3. **System Designer**: Professionals involved in system designing, Logical & Physical Designing wherein logical designing can be enumerated as the structure & characteristics such as output, input, files, database & procedures, etc.

4. **System Analyst**: Computer engineers who work as systems analyst do research about the existing problems and plan solutions of the problem.

5. **Networking Engineers**: Networking engineers are computer professionals involved in designing, implementation, and troubleshooting of computer networks.

6. **DBA**: DBA or Database Administrator is the professionals, who are bestowed with the job to design, implement, maintain, and repair an organization's database.

**Scope of other fields using computer science**

**1) Routine job handling**

The routine classical and stenotype jobs are calculating and formality bits, salaries, updating stocks, tax return, reservation records and information.

**2) Traffic control**

We can control traffic and traffic lights. Television cameras are used to maintain traffic light routine.

**3) Electronic money**

Automatic tellers machine (ATM) is very common in banks. You can deposit and withdraw money with the ATM.

### 4) Electronic office

All type information are stored, manipulated and utilized in the electronic form. A document is sent to different place with FAX, internet and e-mail.

### 5) Industrial Application

It plays an important role in production control. It is bringing efficiency it trade and industry.

### 6) Telephones

With help computerized telephone through satellites STD and IST services have been introduced. It maintains the record of calls and does the billing for you.

### 7) Trade

Every type of trade computer is used successfully. It is used in Banks, stock exchanges to control stocks and accounts.

### 8) Scientific research

In every science, the research work becomes economical from time, energy, money point of new. A large data is analyzed very quickly.

### 9) Medicine

There is wide use in medical science e. g. ECG, CAT scan, Ultra sound. The proper and accounts diagnosis is done with the help of computer. The medical apparatus are controlling computerized.

**10) Space Science**

The satellite controlling I the space with the help of computer. The information's are collected by using the computer from the space satellite.

**11) Defence**

Missiles, weapons, communication employees computerized control system. Success of any military action depends on computers and its applications.

**12) Communications**

The computer is used for sending message example printer, FAX, e-mail, Internet. The import and export work is done on internet.

**13) Film industry**

It had influenced film industry such as animation; tittering etc. The multimedia approach is used in film production with the help of computer. The cartoon films are developed by computers.

**14) Education**

The computer is widely used in the field of education and independent study field of computer science has developed which is popular these days. At every stage computer is compulsory. The distance education is using computer for instructional purpose as multimedia approach. The computer makes teacher learning process effecting by involving audio and visual sense of learners.

### 15) Visual Communication

Visual Communication is the conveyance of ideas and information in forms that can be seen with the help of Computers. Besides two dimentional images, there are other ways to express information visually – body language, animation, graphic designs and film. Advertisement on television, online or on mobile develop by computers.

## 1.4 Need and significance of teaching Computer Science

Computer Science knowledge is very important for other subjects. The aims and objectives of Computer Science teaching have undergone an immense change in the computer era. In the growth and development of an individual and to apply computer knowledge and skills in the working place, we need computer Science knowledge. Realizing the significance of computer science, the teachers should take all out efforts in making the subject more interesting and attractive to the students.

### Need of teaching Computer Science

The arguments for computer science education fall into two main categories: Learning essential tools for the internet age, and learning to understand the world by understanding media. The former category is not about learning specific programs; one should not to decide the value of a course based a programming language. The main skill is learning how to think about process.

Computer Science at its core is largely about the formalization of process. This way of thinking has caused massive changes in areas like operations research, economics, and even mathematics: the discipline from which computer science emerged. I'm hoping to help it also shape more thinking about education, and other disciplines where it has untapped potential. The key insight is that learning computer science will teach you how to present thoughts, ideas or instructions using structure as a tool to avoid ambiguity. This sounds simple, but it's valuable, try and remembers the last time you dealt with ambiguous instructions and how much time you wasted because of that lack of clarity. Computer science helps one learn to organize thoughts to avoid ambiguity; this also helps recognize ambiguities to clarify them. For many, that saved time and gain in communication skills alone is worth taking a university half course, but it remains to address the value of media.

- Everytime, everyone, everywhere, we find the use of computers in some from or other.
- Computer Science is the hall mark of human civilization.
- Only after the advent of computers the rate of knowledge explosion has increased exponentially.
- Computers have been invented by sheer necessity and continued research and how research itself could not be efficiently undertaken without computers.
- The whole world has become global-village because of computers and their need and latest applications.
- To know the world around us.
- To enable the individual to meet the demands of daily life.
- To develop skill in the use of network, internet, social media.
- To understand the culture, social development and civilization.
- To understand the natural phenomena clearly.

- To develop self-confidence, self-reliance and patience.
- To apply mathematical concepts and principles in a new situations.
- To apply computer science knowledge and skills in the working place.
- To apply computer science knowledge in the other school subjects.
- To enable the pupils to appreciate the subject of computer science in all fields.
- To find the alternate methods while solving the problem.
- To develop the skill to express computer science concepts thoughts clearly and accurately.

**Significance of teaching computer science**

Technology has struggled to find its way into the classroom in all sorts of ways, from projectors and televisions to computer labs and student laptops. Along with improving the way students are taught, it is also vitally important that students learn to use computers to improve their own work and prepare for careers in a world where computers have become as common as the pencil and paper.

**1. Modernizing Education**

Education has benefited from the inclusion of technology and computers by making it easier for students to keep up while helping teachers by improving the way lessons can be planned and taught. Students who use computers learn to use word processors for work, and subsequently they learn computer jargon and strengthen grammatical skills. Students can also look up lessons on websites or through email rather than lugging heavy textbooks with them every day.

## 2. Improving Student Performance

Students who use computers have been shown to attend school more steadily and perform better than students who do not use computers. Along with getting higher grades on exams, students also stated they felt more involved with their lessons and work if they used a computer. Using computers gets students to become more focused on their work at home, in collaborative projects with other students and on their own.

## 3. Learning Job Skills

Computers play a vital role in the modern business world, and many of even the most basic jobs involve technology and computers. Teaching students how to use computers helps them prepare for any number of possible careers, and classes based on computer education can get even more specific. Many classes teach students to use office suite programs, create presentations and data sheets, and learn any number of programming languages such as C++ or Java.

## 4. Efficiency

Computers make the learning process a lot more simple and efficient, giving student's access to tools and methods of communication unavailable offline. For example, students can check their grades or lesson plans online and also communicate directly with their teachers via email or educational platforms such as Blackboard. Students can also send work to their teachers from home or anywhere else, letting them finish work outside the constraints of school hours and teaching them about procrastination and personal responsibility.

## 5. Research

Technology has made research far easier than in the past. Decades ago, students learned history by going to the library and thumbing through history books and encyclopedias. Today, many of those same books are available in digital format and can be accessed online. As the Internet has grown, many research options are available. Students can research topics in minutes rather than the hours it used to take.

## 1.5 Values of teaching Computer Science

### Importance and functions of values

Values are general principles to regulate our day to day behaviour. They are not only gives direction to our behaviour but also ideals and objectives in themselves. Values deal what ought to be. Our values are the basis of our judgements about what is desirable, proper, correct, important, worth wile and good. It controls our disruptive behaviours; values enable individuals to feel that they are above us. Live with the following objective.

"Let Noble thoughts come to me from all directions"

### Values of teaching computer science

The real importance of the subject in our modern life can be quite obvious from its chief values which are described below.

### 1. Practical Value

Utilization of the various facts drawn from the study of computer science in modern life has revolutionized our life. Today we cannot find even a single thing which is left untouched by the hands of computer. Uses of computers in transportation and communication have shortened the world.

In short, computers have become a part and parcel of our life and without them, it is impossible for us to keep ourselves alive in the modern world.

## 2. Social Value

Computers have achieved the best place in the society as well. They form the foundations of so many professions like medicine, Engineers etc. Computers are highly helpful to the society. Lots and lots of social changes have taken place after the introduction computers. The study of computer science develops in us honesty, truthfulness and critical reasoning, objective thinking and a belief in basic facts.

## 3. Disciplinary Value

The learning of computer science involves some scientific disciplines and scientific attitudes which are transferable to our later life also. It involves self-expression, creativeness open mindedness, critical thinking and observation suspended judgment.

The Minnesota Educational Computing Consortium (MECC) recently received funding from the National Science Foundation to design, develop, test and evaluate an integrated set of twenty-five student learning modules for science, mathematics and social studies courses in secondary schools. The central theme of these learning packages will be the computer—how it works, how it is used, its impact on the individual and society, and the implications of its use for the future.

The project presumes that abilities, skills, attitudes, and cognitive learning are all aspects of a person's computer literacy; the learning packages will be based on a set of learning objectives that reflect these areas. The materials will be written for students entering junior high school; however, they could be used at other secondary school levels as well. The initial period of the project consists of planning and the specification of objectives which are free from superstitious and false beliefs etc. These good habits if they are once developed in a child can prove beneficial for his later life also.

### 4. Cultural Value

The role of computers in the development of modern civilization can be quite obvious just by our comparison with our ancestors. Our present culture and advancement in our standard of living gives a clear cut picture of our cultural development and role of computers in this field for removing old traditional beliefs and superstitions. Computers have proved it as in best helper in overhauling the consciousness of the universe.

So at the end, from the above mentioned values of computer science, we come to this conclusion that computer science has achieved a most important place in our daily life as well as in the modern establishment of the whole world.

### 5. Moral Values

Moral values are the standards of good and evil which governs an individuals behaviour. Science has more moral values. Science is the search for truth in a faithful manner. Promotion of science along with the growth of moral values in necessary for human development. Scientists strongly abhor fraud, error and pseudoscience while they value reliability, testability, accuracy, precision and simplicity. These values are generally derived from our experience in research Good views of others are accepted by scientists. For the welfare of mankind scientists conduct research and invent new things. It teaches the pupil to be intellectually honest and truthful in our daily activities.

## 1.6 Aims and objectives of teaching Computer Science in schools

### Aims:

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the uses of teaching. We use the term 'aims' to mean the broad goals which our educational system (the whole of the Indian society involved) embraces and which we are expected to attain. The aims of education are based on philosophical and socio-psychological aspects of society and culture. Some of the aims are as follows.

- Education to develop patriotism and good citizenship.
- Education for moral, ethical and spiritual values.
- Education to fulfill the vocational needs of the individual and society.
- Educational for leisure-time activities such as music, dancing, gardening, reading, hobbies, etc.,
- Education for improved health knowledge (physical and mental) and practices.
- Special education for the disabled.

**Objectives:**

Objectives are the milestones to reach the destination i.e. to attain the aim or ultimate goal of education. Objectives are specific, direct and practical in nature; objectives are related to the learning outcomes or change in the behaviour of the students. The objectives are meant for both the teacher and the students. Even at the planning stage i.e. before entering the classroom the teacher asks himself, "what changes in behaviour of the students can I bring about through this lesson?" such changes constitute his (teacher's) educational objectives.

An objectives is thus

- A point or an end-view of something towards which action is directed;
- A planned change sought through an activity

- What we set out to do.

Educational objectives, therefore, consist of the changes we wish to produce in the child. The changes that take place through education can be represented in

- The knowledge children acquire (knowledge)
- The skills and abilities children attain (skill)
- The interest children develop (interest) and
- The attitude children manifest (attitude)

He would understand something which he did not understand before. He can solve problems which he could not solve before. He favorably revises his attitudes towards things. Education is thus the process of bringing about changes in the behaviour of the students in

- What they know (knowledge extends)
- How they think (process of thinking changes)
- How they feel (process of feeling changes)
- How they work (methods of working change)

**Aims of teaching Computer Science in schools**

1. Concepts, information and skills which could be immediately used and applied.
2. Training for developing skills in handling effectively equipment and tools and also in the skill of problem solving.
3. Develop computer science related attitudes.
4. Appreciating the contribution of computer scientists and their inventions.

5. Training in making best use of leisure time and achieve knowledge and skills required for future life and progress.

**Objectives of teaching Computer Science in schools**

**Are Successful Professionals**

- Obtain employment in the computing field
- Gain acceptance into graduate school

**Have a broad knowledge**

- A broad knowledge of the theoretical foundations of computer science
- Knowledge of the fundamental areas of computer science ( programming languages, operating system, computer architecture, programming and software engineering)
- Substantial knowledge of one key area of computer science

**Think Independently, Acquire knowledge and continue their development**

- Acquire new knowledge
- Think independently and rigorously
- Develop a plan for professional development

**Apply Scientific and Engineering Methodology to the Design, Implementation, Analysis and Evaluation**

- Effectively apply scientific and engineering principles to the design of computer based systems
- Implementation of computer based systems
- Analysis of computer based systems
- Evaluation of computer based systems

**Communicate effectively both orally and in writing and collaborate effectively in teams**

- Conduct effective professional technical and non-technical conversations with co-workers and clients
- Communicate effectively in writing
- Work effectively in a team environment with people from other disciplines, throughout the problem solving process

**Are prepared for the Ethical, Societal and Global Issues Associated with the computing field**

- Understand contemporary legal, social and ethical issues in computing
- Identify and analyse legal, social and ethical impacts of professional behaviour and actions

### 1.6.1 Aims and objectives of teaching basic computer science in schools in different stages

#### 1. High School Stage

The following are some of the chief aims and objectives of teaching basic computer science at the High School level. They are

i. Arousing and maintaining interest

ii. Developing the habits of observations exploration, classification and a systematic way of thinking

iii. Developing the child's power of manipulation and so on.

iv. Developing the ability to reach generalizations and to apply them for solving everyday problems using computers.

v. Understanding the Impact of computer science on our way of life.

vi. Developing interest in hobbies related to computer's their generations and so on.

**2. Higher Secondary Stage**

i. To familiarize the pupil with the world in which he is living and to make him understand the Impact of computer science on society so as to enable him to adjust himself to the environments.

ii. To develop scientific attitude this includes

(a) A desire for accurate knowledge

(b) Belief in cause and effect

(c) Critical thinking

(d) Intellectual honesty opens mindedness and so on.

iii. To give the students a historical perspectives, so that they may understand the evolution of computers and development in computers etc.

iv. A broad knowledge of the fundamental areas of Computer science.

v. Obtain employment in the computer field.

vi. Develop a plan for Professional Development.

### 1.6.2. Difference between Aims and Objectives

| S. No. | Aims | Objectives |
|---|---|---|
| 1 | Aims are long term goals | Objectives are short term goals |

| S. No. | Aims | Objectives |
|---|---|---|
| 2 | Aims are broad and common | Objectives are narrow and specific |
| 3 | Philosophy, Sociology is main source of aim | Psychology is the main source of objectives |
| 4 | They are difficult to achieve | they can be achieved conveniently |
| 5 | These can't be evaluated | They can be evaluated |
| 6 | School, Society and Nation is responsible to achieve them | Generally teacher is only responsible |
| 7 | Aims are non-observable | Objectives are observable behaviours |
| 8 | To achieve aims we have to go beyond curriculum | Objectives are achievable within the curriculum |

## 1.7 Instructional Objectives and Behavioural Objectives

### Instructional Objectives

Instructional objectives are basically statement which clearly describe an anticipated learning outcome.

Instructional objectives are helpful to the teacher as well as the learner throughout the learning process and are invaluable in the evaluation process. An instructional objective is the focal point of a lesson plan. Before start teaching teacher must be guided by instructional objectives. Instructional objectives can be classified into three domains of learning i.e. cognitive, affective and psychomotor. Objectives should state what the learners should be able to do after the instruction is complete. The process of how the instruction happens is not considered as an objective.

**Purpose of Writing Instructional Objectives**

- Define the desired outcome of instruction in terms of tasks that student will be able to perform.
- Establish expectation for the student.
- Focus course development and teaching activities on relevant information and skills.
- Provide a clear reason for teaching
- Provide criteria for eliminating unnecessary information and activities
- Indicate how success can be measured

**Characteristics of Instructional objectives**

Instructional objectives are specific, measurable, short term, observable student behavior. They indicate the desirable knowledge, skills of attitudes to be gained at the end of teaching. It describe the performance of a learner should be able to demonstrate after instruction.

- A well-written objective should describe a learning outcome. It should not describe a learning activity
- An objective focuses on the learner, not on the teacher. It describes what the learner will be expected to be able to do. It should not describe a teacher activity.
- If an instructional objective is not observable, it leads to unclear expectations and it will be difficult to determine. The key to writing observable objectives is to use verbs that are observable and lead to a well defined product. Verbs such as "to know" to understand, "to enjoy" to appreciate etc. are vague

and not observable. Verbs such as "to identify", "to list", "to compute" are explicit and describe observable actions or actions that lead to observable products.

- There are many skills that cannot be directly observed. The thinking process of a student to solve a mathematics problem cannot be easily observed. However one can look at the answers or the steps to arrive at an answer.
- Objective to be sequentially appropriate
- Objective to be attainable within reasonable amount of time.
- Objective to be developmentally appropriate (within students level of intellectual)

**Importance of Instructional Objectives**

Instructional objectives are important to both learners and instructors. They help learners plan their study and prepare for examinations. They guide the instructors in planning instruction and devising tests. Instructional objectives play an important role in the teaching and learning process. They are the heart of teaching.

**Importance of Instructional Objectives**

1. To provide direction for the instructional process by clearly stating the intended learning outcomes.
2. To convey instructional intent to students, parents and educational authorities.
3. To provide a basis for evaluating student's learning by describing the performance to be measured.
4. They state in clear terms behavioural changes expected from a learner after with a learning activity.

5. Without instructional objectives, it will be very difficult for a teacher to access the behaviours of the students in the three domains of education. (Cognitive, affective and Psychomotor Domains)
6. Instructional objectives encourage reflection and good course design and development.
7. Instructional objectives help to define reinforcement situations of the learners.
8. They enable the teacher to modify methods to suit the process of eliciting those behaviours that the objectives specify.
9. They help the teacher to know how well the students have covered the content.
10. They give direction to both teacher and students in the selection and use of materials, methods and activities for learning.

**Behavioral objectives**

A behavioral objective is a learning outcome stated in measurable terms, which gives direction to the learner's experience and becomes the basis for student evaluation.

**Points in writing behavioral objectives**

1. Begin each behavioural objectives with a verb. The critical aspect of any behavior is the verb selected from learning activities.
2. State each objective in terms of learner performance. A behavioral objective is one that is considered to be observable and measurable. Behaviour is generally constructed to be an action of an individual that can be seen, felt or heard by another person.

3. State each objective so that it includes only one general learning outcome.

**Examples of Behavioral Objectives**

| | | |
|---|---|---|
| Knowledge | : | Remembering previously learned facts |
| Comprehension | : | Ability to understand or grasp the meaning of material. |
| Application | : | Ability to use previously learned material in new and concrete situations. |

**Motivation and Behavioural Learning**

Motivation plays an important role in behavioural learning. Positive reinforcement can be motivators for students. For example, a student who receive praise for a good test score is much more likely to learn the answers effectively than a student who receives no praise for a good test score. The student who receives no praise is experiencing negative reinforcement - their brains tell them that though they got a good grade, it did not really matter, so the material of the test becomes unimportant to them. Conversely students who receive positive reinforcement see a direct correlation based on that response to a positive stimulus.

**Purpose of Behavioural objectives**

- Guide for the teacher to design of instruction.
- Guide for the teacher for evaluation
- Guide for the learner relative to learning focus
- Guide for the learner relative to self assessment
- Makes teaching more directed and organized

- Forces teacher to think carefully about what is important.
- Helps teacher to avoid unnecessary repetitions in teaching.
- Provides visibility and accountability of decisions made by teachers and learners
- Helps students make decisions regarding prioritizing
- Helps relationship between teacher and learner.

**Examples of Behavioural objectives**

**General Objectives** : The student will know various aids to teach the subject.

**Specific Objectives** : Student must be able to analyse and interpret the results obtained.

### 1.7.1 Writing Instructional Objectives interms of Behavioural Objectives

A behavioral objective is a way by describing the objectives of a content interms of what the learner should be able to do at the end of the lesson. A behavioural objective must be stated clearly and precisely so that everyone who reads it will know exactly the desired outcome of the content.

In writing a behavioural objective, it is important to include the verb that depicts what the learner is expected to do. i.e. it should be measurable.

**Examples**

1) The students will be able to describe the process of transpiration in plants

Here the students are going to describe at the end of instruction. Here the emphasis is on learners and the learning outcome. Here learning is a change of behavior.

2) To identify proper fractions

   Here the learner and his performance are given.

3) The students will understand the three laws of reflection

Here the verb understand does not indicate behavior that can be observed directly. It does not indicate any specific behavior. if the objective is written as.

The students will tell the three laws of reflection.

Then it is observable behavior. Hence the verb 'understand' can further be explained and classified by listing under that some specific behavior like tell, identify, describe, give, differentiate etc. In other words the verb understand is more general and the verb tell is more specific as compared to other.

**Let us discuss the two objective in details**

i. The pupil will know the working of a steam engine.

ii. The pupil will be able to describe in about ten to fifteen sentences the working of a steam engine.

In the above example i) one cannot see other's mind to determine what he knows. We are not able to observe some aspect of this behavior that corresponds his knowledge. Hence an objective should indicate the kind of performance which will be accept as evidence.

Thus in the second example, well-stated instructional objectives are given in observable behavioral terms.

Few examples of well-stated Instructional Objectives

1. The students will be able to describe the three causes of World War II as discussed in the class.
2. The class will be able to identify correctly, all the parts of the hibiscus flower.

3. Students will tell the name of any four corporations in Tamil Nadu.

4. Students will be able to identify the part of a simple sentence.

5. Students draw a right angled triangle with given two sides.

## 1.8 Bloom's Taxonomy of educational objectives

To define understand and classify the educational objectives American educationalist Benjamin S. Bloom and his associates at the university of Chicago divide it into three domains i.e. cognitive, affective and psycho-motor and explains clearly to the teachers.

A brief account about these is given below.

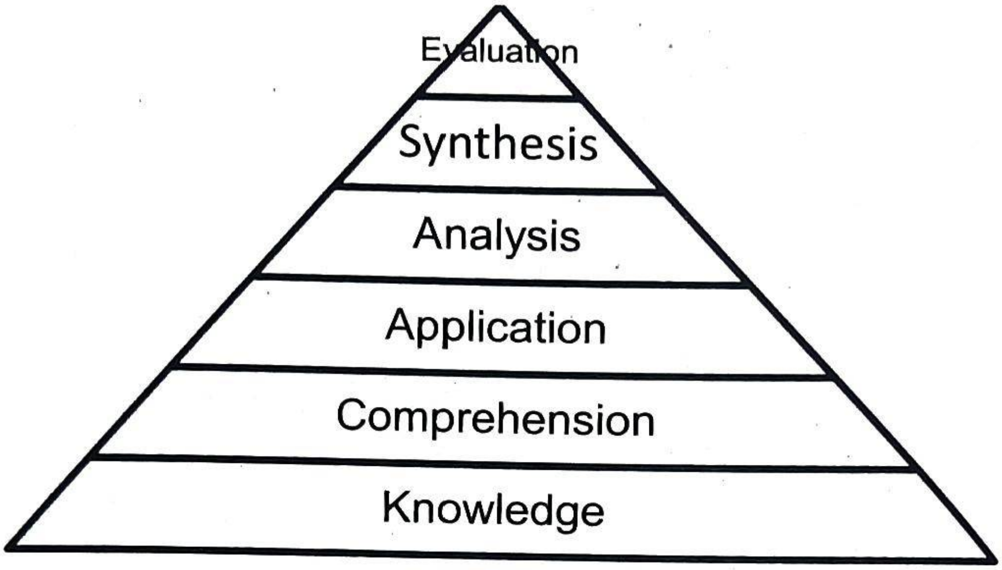
Educational Objectives

| Cognitive Domain (Recall of knowledge and the development of intellectual abilities and skills) | Affective Domain (Development of interests values and adjustments) | Psychomotor Domain (Development of motor Skills) |
|---|---|---|

In each domain, objectives are hierarchical order starting from easy to difficult. Hence this arrangement is called "Taxonomy of Educational objectives.

### 1.8.1. Cognitive domain

Cognitive domain includes those objectives which deal the recall or recognition of knowledge and the development of intellectual abilities and skills. This domain consists of six parts and arranged in a hierarchical order, in increasing levels of difficulty (from lower to upper)




Knowledge forms the basis for all cognitive objectives and all other objectives are not possible without knowledge. Similarly if one is able to apply knowledge in a new situations, it can be inferred that he has to achieve lower levels of objectives namely knowledge and comprehension clearly. Thus a higher level objective subsumes to lower ones.

While teaching Computer Science for Secondary and Higher Secondary students give priority to the first three objectives of cognitive domain namely knowledge, comprehension and application. The remaining three objectives Analysis, Synthesis and Evaluation may be trained to them according to their maturity of mind.

**Knowledge**

This objective is to remember and recall or recognize the already learnt information, acquire more and more knowledge.

**Comprehension**

Objective at this level requires understanding the knowledge without any doubt. He should have ability to arrange data in order.

## Application

This objective requires students to apply already learnt knowledge learnt knowledge to new and unfamiliar situations. Think critically in depth.

## Analysis

Objective at this level requires the learners, to attain the ability to break up a given communication into parts to understand it better.

## Synthesis

Synthesis is just opposite to analysis. The ability of this level is to draw upon elements from various sources and put them together to produce a new structure. This objective gives divergent thinking and creativity.

## Evaluation

This is a highest objective of cognitive domain. The ability of this level is to make judgements about something for given purpose. Judgment can be both quantitative and qualitative.

It helps teachers in the following ways:

- Helps to assess and judge the difficulty of assignments.
- It increase the accuracy of the lesson
- Simplifies action to ease personal learning
- Designed a combined assessment
- Helps for project based plans for learning
- Helps for group discussion

**Instructional objects of Cognitive Doman**

| Parts of Cognitive Domain | Instructional Objectives |
|---|---|
| **Knowledge**<br>Student knows the computer science symbols, concepts, definitions, formulas, statements etc. | **Student**<br>1. Recalls the computer science formulas, concepts and statements.<br>2. Recognises what he learned. |
| **Comprehension**<br>Student understand the of computer science subject symbols, concepts, definitions, formulas, statements | **Student**<br>1. Gives examples<br>2. Give reasons<br>3. Compares<br>4. Differentiates<br>5. classifies<br>6. relates<br>7. detect errors and rectify<br>8. using symbols<br>9. Explains<br>10. Evaluate the result |
| **Application**<br>Using of computer science knowledge and comprehension to solve the problem in a new situation. | **Student**<br>1. Examine what is given in the problem? What is need?<br>2. Examine the sufficiency of the data.<br>3. Select appropriate formula<br>4. Solve the problem in an appropriate method<br>5. Applying alternate method<br>6. Verifying the result. |

| Parts of Cognitive Domain | Instructional Objectives |
|---|---|
| **Analysis**<br>Break up a given of computer science statement into small meaningful parts | **Student**<br>1. Understand the meaning of the hypothesis<br>2. From the given data draw conclusion<br>3. Identify motives / causes for the computer science concepts.<br>4. Discriminates, selects, illustrate computer science concepts. |
| **Synthesis**<br>Putting elements from various sources and put them together to produce a new structure | **Student**<br>1. Design a method to perform a new task.<br>2. Revises the process to improve the outcome.<br>3. Compile, generate, create computer concepts.<br>4. Solves the problem<br>5. Develop the idea in advance. |
| **Evaluation**<br>Evaluating Qualitative or quantitative judgement for computer science elements or methods. | **Student**<br>1. Evaluate the solution of a given problem<br>2. Select the most effective solution.<br>3. Explain and justify a new proposal<br>4. Assess / defend computer science concepts.<br>5. Review / Criticise computer science statements. |

### 1.8.2. Affective domain

The affective domain refers to emotional and attitudinal engagement with the subject matter while the cognitive domain refers to knowledge and intellectual skills related to the subject matter.

Affective domain includes the development of feelings of the educational and social attitudes and to increase the emotional objectives namely attitude, interest, values and appreciation. The different objectives under affective domain are receiving, responding, valuing, organisation and characterization.

#### 1. Receiving

This is the first step for the learner to learn what the teacher teaches. Listen attentively to a particular phenomena or stimuli.

#### 2. Responding

This is also called interest objective. Pupil is sufficiently motivated to attend actively. Interest in computer science lends the student to select a particular vocation throughout his life. Interest can be developed in solving puzzles, reading computer science books and journals etc.

#### 3. Appreciation

Student appreciates computer science for its use in daily life and how it helps in other fields.

### 4. Organisation

This objective brings together different values, resolving conflicts among them, starting to build an internally consistent value system. Explains the role of systematic planning in solving problems.

### 5. Valuing

It means individual's own valuing or assessment of thing, phenomena or behaviour. It is also called attitude. Development of proper attitude in computer science is one of the major objectives of teaching computer science. Proposes a plan to solve social problem.

**Instructional Objectives of Affective Domain**

| Parts of Affective Domain | Instructional Objectives |
|---|---|
| **Receiving**<br>Listen to others with respect. Listen for and remember the newly introduced content. | **Student**<br>1. Identifies<br>2. Asks<br>3. Understand<br>4. Points out errors boldly.<br>5. Listen to others attentively. |
| **Responding**<br>Participate in class discussions. Gives a presentation, questions new ideas, concepts etc. | **Student**<br>1. Reads books of Computer science<br>2. Solves science puzzles<br>3. Writes activities on science for school magazine.<br>4. Gives short cuts for solving problems.<br>5. Does problems apart from syllabus.<br>6. Gains Computer science knowledge more than the syllabus. |

| | |
|---|---|
| | 7. Participates in computer science club activities.<br>8. Participate in class discussions. |
| **Appreciation**<br>Valuing to a particular object, phenomenon or behavior ranges from simple acceptance to the more complex state of commitment. | **Student**<br>1. Solving the problems of other subjects using computer.<br>2. Appreciates how computer is useful in his daily life.<br>3. Sensitive towards individual and cultural differences. |
| **Organisation**<br>Recognise the need for balance between freedom and responsible behavior. Explains the role of systematic planning in solving problems. | **Student**<br>1. Recognize the need for balance between freedom and responsible behavior<br>2. Explain the role of systematic planning in solving the problems.<br>3. It becomes habit to think logically. |
| **Valuing**<br>Has a value system that controls his behavior. Uses an objective approach in problem solving. Revises judgments and changes behavior in the light of new evidence. | **Student**<br>1. Shows self reliance when works independently<br>2. Co-operates in group activities.<br>3. Revises judgments and changes behavior in light of new evidence.<br>4. Values people for what they are not how they look.<br>5. Accepted statements when logically proved. |

### 1.8.3. Psychomotor domain

Elizebath Simpson built her taxonomy on the work of Bloom and others. Simpson's Psychomotor domain is comprised of utilizing motor skills and coordinating them. Simpson's taxonomy has a focus toward the progression of mastery of a skill from observation to invention.

The psychomotor domain includes Physical movement, coordination and use of the motor-skill areas. Development of these skills requires practice and is measured in terms of speed, precision, procedures or techniques in execution. The seven major categories are perception, set, Guided Responses mechanism, complex over to Response, Adaptation and Origination.

The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

**1. Perception**

It means observing the act of the teacher.

**2. Set**

Readiness to act. It includes mental, physical and emotional sets

**3. Guided Response**

Following directions of the teacher and differentiating among various movements. This is the early stage in learning complex skill.

**4. Mechanism**

This is the intermediate step in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency

### 5. Complex Overt Response

This is the highest level of proficiency and it becomes natural and automatic. Proficiency is indicated by a quick, accurate and highly coordinated performance, requiring a minimum energy.

### 6. Adaptation

Skills are well developed and the individual can modify movement patterns to fit special requirements

### 7. Origination

Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.

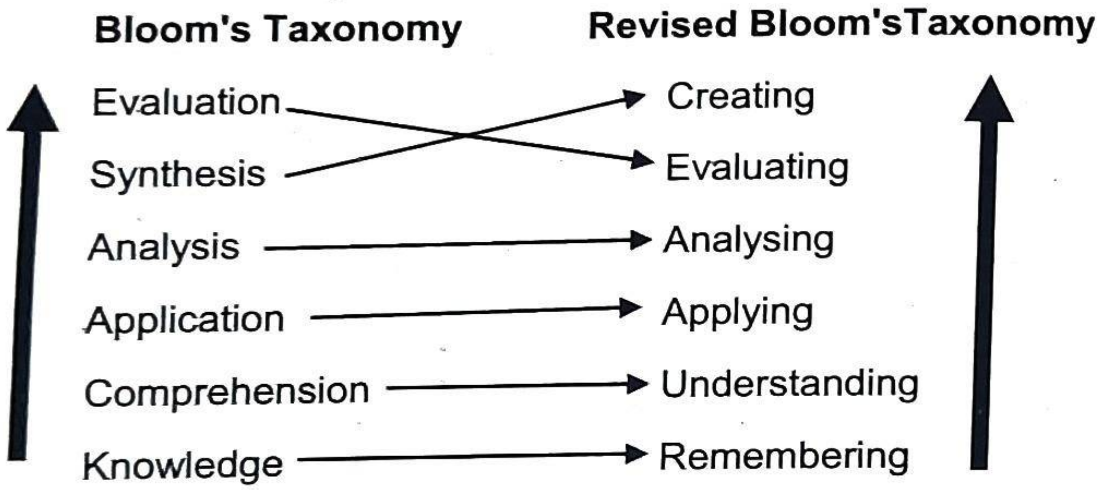
**Instructional Objectives of Psychomotor Domain**

| Parts of Psychomotor Domain | Instructional Objectives |
|---|---|
| **Perception**<br>Perception is the most basic level to process sensory information that guide motor activity | **Student**<br>1. Detects non verbal communication cues<br>2. Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball.<br>3. Observing the act of the teacher. |
| **Set**<br>Mentally, emotionally and physically ready to act. | **Student**<br>1. Repeat the act of the teacher.<br>2. Acts upon a sequence of steps to solve a problem.<br>3. Shows desire to learn a new process. |

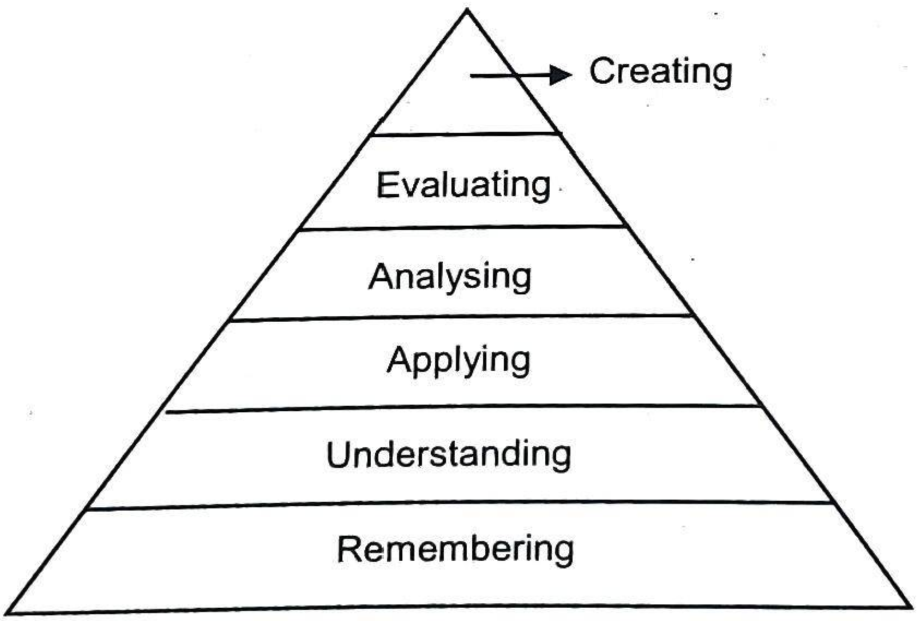
| Parts of Psychomotor Domain | Instructional Objectives |
|---|---|
| **Guided Response**<br>Imitates and practices skills often in discrete steps | **Student**<br>1. follows<br>2. react<br>3. response<br>4. copies<br>5. performs |
| **Mechanism**<br>Performs acts with increasing efficiency, co-ordinate and proficiency. | **Student**<br>1. Drives a cycle<br>2. assembles<br>3. dismantles<br>4. measures<br>5. mixes |
| **Complex Overt Response**<br>Motor activity performs automatically. | **Student**<br>1. Operates a Computer quickly and accurately<br>2. assembles<br>3. builds<br>4. measures<br>5. organizies |
| **Adaptation**<br>Adapts skill sets to meet a problem situation. | **Student**<br>1. Responds effectively to unexpected situations.<br>2. Modifies method to meet the needs.<br>3. Perform a task with a new method. |
| **Origination**<br>Create new patterns for specific situations. | **Student**<br>1. Constructs a new theory.<br>2. designs<br>3. creates<br>4. builds<br>5. composes |

## 1.9 Revised Bloom's Taxonomy

In 1990 Loris Anderson former student of Bloom revised the taxonomy and made a number of changes.




**Revised Bloom's Taxonomy**




**Remembering:**

The knowledge category was renamed as remembering. Knowledge is an outcome or product of thinking and not a form of thinking. Hence the word knowledge was inappropriate to describe a category of thinking and was replaced with the word remembering.

Remembering is recalling information. This objective is to recognizing, listing, describing, naming, finding already learnt information.

**Understanding:**

Objective at this level requires learners to understand the knowledge without any doubt. He should have knowledge to interpret, explain, classify and summarize.

**Applying:**

Objective at this level requires the learners to apply information in another familiar situation implementing and executing in a best way.

**Analysing:**

Objective at this level requires the learners to attain the ability to break up a given information into parts to explore understanding and relationships between them. He should have the knowledge of diffenticiating, organizing and integrating.

**Evaluating:**

The ability to make judgements about a decision or course of action. Judgement can be both quantitative or qualitative. He should have the knowledge of checking, deducting and judgeing.

**Creating:**

By putting elements together from various sources to form the ability to generate new ideas, products or ways of viewing thing in a different way. Designing, planning, producing, inventing new things and objects is the objective at this level.

**Potential activities of cognitive domain**

**Remembering:**

1. Make a list of the main events of today.
2. Write parts of computer.
3. Objective questions are used to measure the learner's skill.
4. Make a chart showing .............
5. Make a time line of events

**Understanding:**

1. Retell the story in your own words.
2. Prepare a flow chart to illustrate the sequence of events.
3. Comparing the concepts
4. Write and perform a play based on the story.
5. Write a summary report of the event.

**Applying:**

1. Construct a model to demonstrate how it works.
2. Prepare an assignment of the topic "Popular Linux distributions".
3. Solving or explaining the problem
4. Make up a puzzle or a game about the topic.
5. Dress a doll in national costume.

**Analyzing:**

1. Design a questionnaire to gather information.
2. Construct a graph to illustrate given information.
3. Here thinking skill is developed
4. Write a Biography of a person studied
5. Make a family tree showing relationships

**Evaluating:**

1. Conduct a debate about an issue of .......
2. Prepare a case to present your view about.......
3. Using different ideas, student understand the concepts
4. Review the Computer Science book for XI Std.
5. Form a panel to discuss views.

**Creating:**

1. Invent a machine to do a specific task.
2. Design a magazine cover for sports day function.
3. Developing skill to create a new hypothesis.
4. Write about your feelings in relation to .................
5. Design a magazine cover for your school annual sports day.

## 1.10 Inter-relation among the Domains

Learning helps develop an individual's attitude as well as encourage the acquisition of new knowledge and new skills.

Learning can generally be categorized into three domains: Cognitive, affective and psychomotor. Learning is not an event. It is a process. It is the continual growth and change in the brain's architecture that results from the many ways. We take in information, process it, connect it, catalogue it and use it.

### Cognitive Development

Cognitive development refers to the way a child grows intellectually. Assessing for academic skills, reasoning, and critical thinking abilities, and the acquisition of knowledge or learning. Difficulties interacting with others as a result of problems and social or emotional development will impede a child's ability to focus. Problems or delays in physical development might impact a child cognitively.

### Affective Domain

The affective domain includes the feelings, emotions and attitudes of the individuals. The categories of affective domain include receiving phenomena; responding to phenomena; valuing; organization and characterization. The sub domain of receiving phenomena creates the awareness of feelings and emotions as well as the ability to utilize selected attention. This can include listening attentively to lessons in class. The next sub domain of responding to phenomena involves active participation of the learner in class or during group discussion. The ability of the student to prioritize a value over another and create a unique value system is known as organization. This can be assessed with the need to value one's academic work as against their social relationships. This helps to control the behavior of the individual.

**Psychomotor domain**

This includes utilizing motor skills and the ability to coordinate them. The sub domains are perception; set; guided response; mechanism; complex overt response; adaptation; and origination. Perception involves the ability to apply sensory information to motor activity. for eg. Practicing a series of exercises for scoring high marks. Set, involves the readiness to act upon a series of challenges to overcome them. It includes the ability to imitate a displayed behavior or utilize a trial and error method to resolve a situation. Mechanism involves the ability to convert learned responses into habitual actions with proficiency and confidence. A student having an increased typing speed when using a computer is an example for complex overt response. Adaptability is an integral part of the domain which exhibits the ability to modify learned skills to meet special events – making a working model. Origination also involves creating new movement patterns for a specific situation.

Cognitive is for mental skills, affective is for growth in feelings or emotional areas while psychomotor is manual or physical skills.

The learning domains can be incorporated, while designing the course outcomes of all the courses in a program. We can learn mental skills, develop our attitudes and acquire new physical skills as we perform the activities of our daily living.

Research in the field establishes the significant relation among three domain of learning.

A lesson developed by a teacher requires the inclusion of all three domains in constructing learning tasks for students. The diversity in such learning tasks helps creates a comparatively well-rounded learning experience that meets a number of learning styles and learning modalities. It is important for teachers to adopt a teaching strategy that combines three domains of learning to enable teaching and learning to be considered as effective. Apply the best and suitable delivery techniques that would impact positively on the Cognitive, Affective and Psychomotor domains of the students. The interaction of all three domains is needed for meaningful educational experiences.

For example which teaching computer science, New knowledge is obtained in Cognitive Domain. Using Psychomotor Domain motor skills, sensory organs (eye, ear) are used while operating a computer. With the Affective Domain components, we give motivation, Acts upon a sequence of steps to go for a process, acts with efficiency and proficiency and creating new technique for specific situations. Teachers has to relate two or three domains and perform teaching learning activities and give meaningful learning experiences.

## 1.11 Correlation between Computer Science and other subjects

People in all walks of life use computers intensively. We all download information to handle our daily life. Computerized equipments are everywhere and computer science is essential in the new era. Information technology is an enormous part of most of the business and the computer industry occupies an important place in the modern world.

### Mathematics and Computer Science

A computer is a device that can person arithmetic or logical operations. Maths matters for computer science because it teaches students how to we abstract language, work with algorithms, self analyse their computational thinking and accurately modelling real-world solutions.

### Engineering and Computer Science

Computer science has strong connect with engineering. Computers are now indispensable when it comes to designing and building any complex structure, from skyscrapers or designing submarine. CAD/CAM systems

(Computer-aided design / Computer-aided manufacturing) rely on a combination of computer graphics and a supporting software to provide the engineer with a set of tools by which one can master the complexity involved.

**Physics and Computer Science**

The two disciplines complement each other, with Physics providing an analytic problem-solving outlook and basic understanding of nature which computer science enhances the ability to make practical and marketable applications. There are also a number of areas at the interface of Computer Science and Physics.

1. Computer hardware is based on solid state physics devices.
2. Large scale simulations
3. Physics of computation (Quantum computing, reversible computing etc.)
4. MRI scans (Magnetic Resonance Imaging) might be impossible without computers.
5. Large no. of physical instruments are used in the health care to diagnose the disease and for treatments.
6. Computer Science helped overcoming difficulties in Physics understanding.
7. Computer Science promoted knowledge integration process.

**Chemistry and Computer Science**

Nobel prize winner karplus said using computers has become a Central part of Chemistry. Computational chemistry allows a computer to understand a specific aspect of science - such as the structure of a protein - and then learn how it functions. Applications of coupling Computers and chemistry include creating solar cells and drugs. Computer science used in Analytical chemistry and Quantum chemistry. Molecules that use in computer are

Silicon

Germanium

Copper

Aluminium

Leed

Mercury

Lithium and more

In laptop lithium-ion battery is used.

**Computer science and Biology**

Computational biology, which is also known as bio informatics is the combined application of maths, statistics and computer science to solve biology based problems. Biology and computer science go hand in hand in the field of Genomics, clinical research, synthetic biology, Bio-technology and Pharmaceutical industry.

**Computer science and History**

We can store lot of historical data on topic wise or on chronological order in computers and to retrieving. History often is a lot of documents. Database can be immensely helpful for storing, accessing and cross-referencing large data and complex algorithms can be used to analyse such data.

**Computer science and language**

Computers are used for storing, indexing and retrieving large collections of written or spoken human language. These are used by lexicographers to write dictionaries and by syntax / language teaching experts to write grammar books. Linguistics and computer science both deal with the recognition and generation of languages.

## REVIEW QUESTIONS

1. Explain the meaning, nature and scope of Computer Science.
2. What are the important Aims of teaching Computer Science in Schools?
3. Write the different values of Computer Science.
4. Explain the importance of Instruction objectives and Behavioural objectives in teaching.
5. Explain the Cognitive Domain and write its Instructional Objectives.
6. Explain the Affective Domain and write its Instructional Objectives.
7. Explain the Psychomotor Domain and write its Instructional Objectives.
8. Explain Revised Bloom's of Taxonomy.
9. How learning domains can be inter-related in teaching of Computer Science.
10. Explain how Computer Science is correlated with other subjects.